Non-Supervised Learning in Action Quality Assessment

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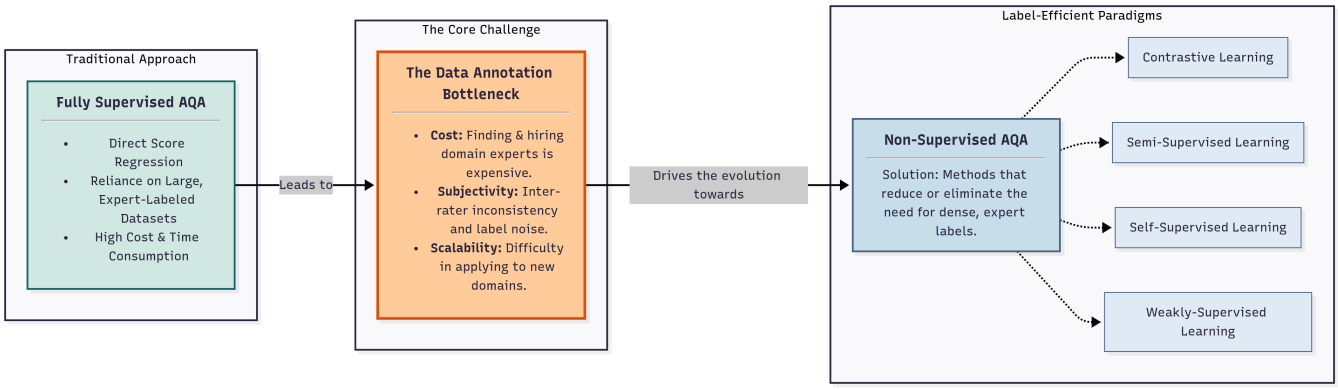


Figure 1: The evolution of Action Quality Assessment (AQA) paradigms, moving from the traditional fully supervised approach and its data annotation bottleneck towards various label-efficient methodologies.

ABSTRACT

Action Quality Assessment (AQA) is a rapidly growing field in computer vision that aims to evaluate not just what action is performed, but how well it is executed. AQA has significant applications in domains like professional sports, healthcare, and industrial robotics, where objective and scalable feedback is crucial. Early and many current approaches have framed AQA as a fully supervised regression task, relying on large datasets with expert-annotated quality scores. However, the creation of these datasets is a major bottleneck, as it is expensive, time-consuming, and often suffers from the subjective bias of human annotators. This dependency on extensive, high-quality labels limits the scalability and applicability of AQA systems.

To address this fundamental challenge, the field is increasingly shifting towards label-efficient and non-supervised methodologies. In this work, we use the term "non-supervised" to cover label-efficient methodologies which aimed at reducing the reliance on densely labeled data. This paper provides a comprehensive survey of this evolution, from traditional supervised methods to the current label-efficient AQA. Label-efficient methods are approaches that try to reduce the need for a large number of high-quality labeled examples. Instead of depending only on expert annotations, these methods use alternative ways to learn from data. We introduce a structured taxonomy to categorize these diverse approaches based on their learning mechanisms and the nature of the supervision they require. Through detailed analysis, we compare these paradigms, highlighting their respective strengths, weaknesses, and performance on key benchmarks. Furthermore, we identify critical open challenges and analyze the emerging paradigms that address them, such as the development of human-in-the-loop systems, hybrid neuro-symbolic architectures for enhanced explainability, and the role of large-scale foundation models. This survey aims to provide researchers and practitioners with a clear roadmap of the AQA landscape and to catalyze the development of more scalable, robust, and equitable AQA systems.

CCS CONCEPTS

• **Computing methodologies** → **Computer vision tasks**; **Unsupervised learning**; **Activity recognition and understanding**; • **Applied computing** → *Life and medical sciences.*

KEYWORDS

Action Quality Assessment, Unsupervised Learning, Self-Supervised Learning, Semi-Supervised Learning, Weakly-Supervised Learning, Video Understanding, Label-Efficient Learning, Non-Supervised Learning

1 INTRODUCTION

The analysis of human actions in video has long been a cornerstone of computer vision research. While traditional action recognition focuses on classifying *what* action is being performed, a growing number of critical applications demand a more nuanced understanding: they require an evaluation of *how well* an action is executed. This task, known as Action Quality Assessment (AQA), has emerged as an important research area with implements in different areas. In domains such as professional sports, AQA systems can provide objective scores for athletes in events like diving or gymnastics [7, 14]. In healthcare, they can assess the proficiency of surgical trainees or monitor the progress of patients undergoing physical rehabilitation [3, 21].

The dominant paradigm for AQA has been fully supervised learning. In this setup, models, typically deep neural networks,

are trained to regress a quality score directly from input video features [8]. These models have achieved impressive results, driven by the development of powerful architectures like 3D Convolutional Neural Networks (3D CNNs) [7] and the creation of large-scale, annotated datasets such as MTL-AQA [7] and FineDiving [14]. These datasets provide thousands of videos with corresponding scores assigned by expert human judges.

However, this reliance on full supervision presents a fundamental and persistent challenge: the data annotation bottleneck. Acquiring high-quality labels for AQA is a time consuming process. It requires domain experts such as Olympic judges or senior surgeons, but the fact is that their time is both expensive and limited. Also, human-assigned scores are inherently subjective, leading to label noise and inter-rater variability, which can degrade model performance and trustworthiness [5, 15]. This "label-dependency" severely limits the scalability of supervised AQA creates a huge problem in its application in new or niche domains where large, annotated datasets are simply not available.

In response to this challenge, the field is undergoing a significant shift towards learning paradigms that reduce or eliminate the need for exhaustive manual labeling. This survey provides a comprehensive overview of this evolution, charting the course from fully supervised methods to a diverse ecosystem of non-supervised and label-efficient techniques. We adopt the term "non-supervised" as an umbrella to encompass a wide spectrum of approaches, including:

- **Contrastive and Relative Learning**, which reframe the task from predicting absolute scores to learning relative differences between pairs of actions.
- **Semi-Supervised Learning**, which leverages a small set of labeled data alongside a much larger pool of unlabeled data.
- **Self-Supervised Learning**, which devises pretext tasks to learn meaningful representations from unlabeled video data itself.
- **Weakly-Supervised and Self-Training Methods**, which utilize other forms of supervision, such as textual descriptions or the model's own predictions, to generate training signals.

The primary contribution of this paper is to synthesize this burgeoning area of research and provide a clear, structured overview for both newcomers and experts. We introduce a novel taxonomy that categorizes these label-efficient methods based on their underlying learning mechanisms and the type of supervision they exploit. By systematically reviewing and comparing these paradigms, we illuminate the trade-offs between performance, label efficiency, and applicability. Finally, we identify critical open challenges—such as model generalization, explainability, and the integration of human-in-the-loop systems—and discuss promising future directions, including the role of foundation models and neuro-symbolic approaches. Through this survey, we aim to provide a clear roadmap of the AQA landscape and to foster the development of the next generation of scalable, robust, and equitable AQA systems.

The remainder of this paper is organized as follows. Section 2 provides a foundational overview of the AQA task, its evolution, and key datasets. Section 3 describes the methodology used to conduct this survey. Section 4 presents our comprehensive survey of learning paradigms, structured according to our proposed taxonomy. Section 5 offers a comparative analysis, discusses open challenges, and outlines future research directions. Finally, Section 6 concludes the paper.

2 BACKGROUND: THE AQA LANDSCAPE

To understand the shift towards non-supervised learning, it is essential to first understanding the fundamentals of the AQA task, the datasets that have driven its progress, and the inherent limitations of the traditional supervised approach.

2.1 The AQA Task: From Regression to Structured Analysis

At its core, AQA is the task of assigning a quantitative score to a video of a performed action that reflects its quality of execution.

2.1.1 AQA as Score Regression. The pioneering work of Pirsiavash et al. [8] formulated AQA as a supervised regression problem. Their approach involved extracting spatio-temporal pose features from videos and training a regression model, such as a Support Vector Regressor (SVR), to map these features to an expert-assigned score. This established a foundational methodology that many subsequent works have built upon, progressively replacing handcrafted features with more powerful deep learning representations from 3D CNNs like C3D [6] and I3D [4].

2.1.2 Structured and Multitask AQA. As the field matured, researchers recognized that a single, holistic score often fails to capture the multifaceted nature of complex actions. This led to the development of more structured approaches. Parmar and Morris [7] introduced a multitask learning (MTL) framework that simultaneously predicts an AQA score, performs action recognition, and generates textual commentary. This approach demonstrated that forcing a model to learn related auxiliary tasks can lead to more robust and generalized representations for AQA.

2.1.3 Procedure-Aware and Group AQA. Further works came from recognizing that many actions, particularly in sports, are composed of distinct phases or sub-actions. Xu et al. [14] proposed the concept of "procedure-aware" AQA, where a model first temporally segments an action (e.g., a dive into take-off, flight, and entry phases) and then assesses the quality of each segment. This fine-grained analysis more closely mimics how human experts evaluate performances and improves both accuracy and interpretability. It also provides an important opportunity to improve understanding and explanation through sub-section scoring rather than using only a general score. More recently, the scope of AQA has expanded to multi-person scenarios. Zhang et al. [19] tackled group AQA in the context of artistic swimming, introducing a group-aware attention module to model the complex interactions and formations among multiple athletes over long-form videos.

2.2 Foundational Datasets and Metrics

The progress in AQA has been linked to the availability of public datasets. These benchmarks not only enable the training of models but also provide a common ground for comparing different approaches. Table 1 summarizes several key AQA datasets cited in the literature.

Table 1: A summary of key Action Quality Assessment datasets. The table highlights the diversity in tasks, annotation richness, and scale, underscoring the significant effort required for data collection in this domain.

Dataset	Primary Task(s)	# Videos	Avg. Dur.	Annotations	Key Reference
MIT Dive	Diving	159	6.0s	Scores	Pirsiavash et al. [8]
AQA-7	7 Sports	~1,106	4.1s	Scores	Yu et al. [15]
MTL-AQA	Diving	1,412	4.1s	Scores, Action Classes, Commentary	Parmar & Morris [7]
FineDiving	Diving	3,000	4.2s	Scores, Temporal Segments	Xu et al. [14]
JIGSAWS	Surgical Tasks	103	~2 min	Scores, Kinematics, Sub-tasks	Gao et al. [2]
Rhythmic Gymnastics	Gymnastics	1,000	95s	Scores	Zeng et al. [17]
LOGO	Artistic Swimming	200	204.2s	Scores, Formations, Action Labels	Zhang et al. [19]

Early datasets like MIT Dive [8] and UNLV-Dive [6] focused on a single activity. The AQA-7 dataset [15] expanded this to include seven different sports, providing a benchmark for generalization. The MTL-AQA dataset [7] was a significant step forward, offering not just scores but also action class labels and textual commentary, enabling the development of richer, multitask models.

More recent datasets have focused on increasing granularity and complexity. The FineDiving dataset [14] provides fine-grained temporal annotations for sub-actions, enabling procedure-aware AQA. The LOGO dataset [19] addresses the challenge of long-form, multi-person AQA with videos averaging over three minutes and containing detailed annotations of group formations. In non-sport domains, JIGSAWS [2] is a key resource for surgical skill assessment, providing synchronized video and kinematic data.

To evaluate performance, the most common metric is the **Spearman's Rank Correlation Coefficient (ρ)**. This metric measures the monotonic relationship between the predicted scores and the ground-truth scores, effectively evaluating how well a model can rank performances. A higher ρ indicates better ranking ability. For a more direct measure of prediction error, some works also report the **Relative L2-distance (R-l2)** [15], which normalizes the mean squared error by the score range of the specific action.

2.3 The Challenge of Full Supervision

Despite the progress driven by these datasets, their creation highlights the central limitation of the supervised paradigm. Especially as noted by Xu et al. [14], the annotation of the FineDiving dataset required six professional annotators and approximately 120 hours of work. This process is not only labor-intensive but also comes with many challenges:

- **Cost and Expertise:** Qualified AQA datasets requires domain experts who are both difficult to access and expensive to hire.
- **Subjectivity and Noise:** Human judgment is subjective. Different experts may assign different scores to the same performance, introducing label noise and bias into the training data [15, 18]. Tang et al. [12] specifically address this by giving a method to learn a score distribution rather than a single point estimate, with a way accepting label noise and bias.
- **Scalability:** A huge effort is required to create large-scale, high-quality AQA datasets; but makes it very difficult to apply the supervised paradigm to every new domain or action of interest.

These challenges provide a clear and huge motivation for exploring alternative learning strategies that can build effective AQA systems with significantly less reliance on manual, expert-driven annotation on seek label-efficient methodologies.

3 METHODOLOGY OF THE SURVEY

This survey aims to provide a structured and comprehensive overview of the transition from fully supervised to non-supervised and label-efficient learning in Action Quality Assessment. Our methodology is guided by the principles of systematic review and conceptual analysis.

3.1 Literature Selection

The body of literature for this survey was compiled from source papers provided. The literature for this survey was compiled through searches in major databases (IEEE Xplore, ACM Digital Library, and arXiv) and cross-references from key AQA publications. We included works published between 2014 and 2025, covering the period since the first formalization of AQA as a supervised regression problem. Papers were selected if they (i) directly addressed AQA or closely related tasks such as skill assessment, (ii) reported quantitative results on established AQA benchmarks (e.g., AQA-7, MTL-AQA, FineDiving, JIGSAWS), or (iii) introduced new paradigms relevant to label-efficient learning. Both peer-reviewed works and influential preprints were considered, but we clearly distinguish between them where relevant. This procedure ensures that our survey captures the breadth of the field while maintaining transparency in inclusion criteria.

3.2 Taxonomy and Structure

A core contribution of this survey is a novel taxonomy for organizing the diverse methods in AQA. Simple chronological or alphabetical listings are insufficient for revealing the underlying conceptual shifts in the field. Instead, our taxonomy is structured around the central theme of this survey: the **type and amount of supervision** required by the learning algorithm.

The taxonomy is structured around the degree of supervision required, with label-efficient paradigms forming the central organizing principle. Categories such as semi-supervised, self-supervised, and weakly-supervised learning were not arbitrarily defined, but directly adopted from the terminology used in the original publications. By aligning with the authors' own categorizations, while placing them under the broader label-efficient umbrella, we ensure both fidelity to the source literature and conceptual clarity in presenting the evolutionary trajectory of AQA methods. The major categories in our survey, detailed in Section 4, are:

(1) Fully Supervised Paradigms
(2) A Taxonomy of Label-Efficient Learning
(3) Contrastive and Relative Learning
(4) Semi-Supervised Learning
(5) Self-Supervised and Unsupervised Learning
(6) Vision-Language and External Knowledge Models

Within each category, we analyze the key ideas, representative methods, and contributions of the relevant source papers. The category names used in this classification come from the terminology that the relevant studies use to define themselves directly. This structure facilitates a clear comparison of how different methods approach the problem of reducing supervision while maintaining or improving AQA performance.

4 A SURVEY OF LEARNING PARADIGMS IN AQA

In this section, we provide an overview of the different learning methods in Action Quality Assessment, organizing them by the amount of supervision they need. We show how the field has evolved from its early supervised approaches to the modern methods that are more label-efficient.

4.1 Fully Supervised Paradigms

The foundation of AQA is built on fully supervised learning, where models are trained on datasets with explicit, ground-truth quality scores for every video.

4.1.1 Direct Score Regression. The initial and most direct approach to AQA is to frame it as a regression problem. Given a video, the goal is to predict a single, continuous score. Pirsiavash et al. [8] pioneered this approach by extracting spatio-temporal pose features and using a Support Vector Regressor (SVR) to predict scores for Olympic diving. Subsequent works replaced these feature engineering steps with end-to-end deep learning models. Parmar and Morris [6] demonstrated the effectiveness of using features from a 3D CNN (C3D) pre-trained for action recognition, feeding them into an SVR or LSTM to regress the final score. Although these approaches successfully established that quality scores can be learned directly from video, their fundamental dependence on a complete set of expert-labeled examples highlights a significant bottleneck, underscoring the need for more label-efficient methods.

4.1.2 Structured and Multitask Learning. Recognizing that a single score is a coarse representation of a complex performance, researchers began to incorporate more structure into supervised models. The work by Parmar and Morris on multitask learning (MTL-AQA) [7] was a significant step in this direction. Their model was trained not only to predict a score but also to perform action classification and generate textual commentary. The underlying hypothesis, supported by their results, is that these auxiliary tasks act as a form of regularization, forcing the model to learn more robust and meaningful features that are beneficial for the primary AQA task.

4.2 A Taxonomy of Label-Efficient Learning

To navigate the diverse landscape of methods that move beyond full supervision, it is helpful to establish a clear taxonomy. These methods exist on a spectrum, defined by the nature and amount of labeled data they require. Table 2 provides a comparative overview.

- **Contrastive Learning** in AQA focuses on learning from the relationship between videos by comparing them. Specifically, approaches like contrastive regression train a model to predict the precise score difference between a pair of videos, rather than predicting a score from a single video in isolation. This requires supervision in the form of absolute scores for each training video, which are used to calculate the relative score targets that the model learns to regress.
- **Semi-Supervised Learning (SemSL)** is a direct approach to label scarcity, explicitly combining a small labeled set with a large unlabeled one. Its prerequisite is the availability of both data types for the same domain.
- **Self-Supervised Learning (SelSL)** is technically a subset of unsupervised learning where no manual labels are used in the representation learning phase. It requires a large corpus of unlabeled data and, critically, a well-designed pretext task that forces the model to learn features relevant to the downstream AQA task.
- **Weakly-Supervised Learning** is a broad category where the supervision is not in the form of dense, exact labels. This can include video-level action tags, textual descriptions, or the noisy pseudo-labels generated during self-training.

These paradigms are not mutually exclusive; for instance, contrastive methods are often used as the learning objective in self-supervised pretext tasks.

4.3 Contrastive and Relative Learning

A key insight that prompted a shift away from direct, absolute score regression was the realization that learning relative quality differences is often more robust. This paradigm, known as contrastive regression, was effectively introduced to AQA by Yu et al. with their framework, CoRe (Contrastive Regression) [15].

Instead of predicting a score for each video, the CoRe framework takes pairs of videos as input and is trained to predict the difference in their scores. By training the model to distinguish between high-quality and low-quality actions in a pairwise manner, CoRe learns a quality-aware representation space. This approach is designed to alleviate the impact of subjective scoring from human judges, as the learning signal is based on relative differences rather than absolute values. The work of Qi et al. [9] further refines this concept by incorporating hierarchical pose information and multi-stage segmentation into a contrastive regression framework. Their research demonstrates that combining structural awareness from poses with relative learning yields significant performance gains.

Table 2: A taxonomy of label-efficient learning paradigms in AQA, comparing their core principles and data requirements.

Paradigm	Core Principle	Data Requirements	Example Application
Contrastive	Learns by comparing pairs of data points to predict their relative quality.	Requires a dataset where each video has a precise, absolute score. The model then computes the score difference between pairs of videos and learns to predict this difference.	Predicting a precise score for a dive by learning its quality relative to a reference (exemplar) video that has a known score. This approach aims to both rank videos and predict their accurate numerical scores [15].
Semi-Supervised	Uses a small set of labeled data to guide learning on a large set of unlabeled data.	A small, fully labeled dataset and a large, unlabeled dataset.	Training an AQA model with a few expert-scored videos and thousands of unlabeled internet videos [16].
Self-Supervised	Learns representations by solving a "pretext" task where labels are generated from the data itself.	A large, unlabeled dataset. Domain knowledge is often used to design the pretext task.	Learning workout form from unlabeled gym videos by predicting temporal order or pose similarity [5].
Weakly-Supervised	Learns from noisy, incomplete, or inexact labels instead of dense, ground-truth scores.	Data with weak labels (e.g., video-level tags, textual descriptions, self-generated pseudo-labels).	Using textual commentary from sports broadcasts to supervise an AQA model [7, 22].

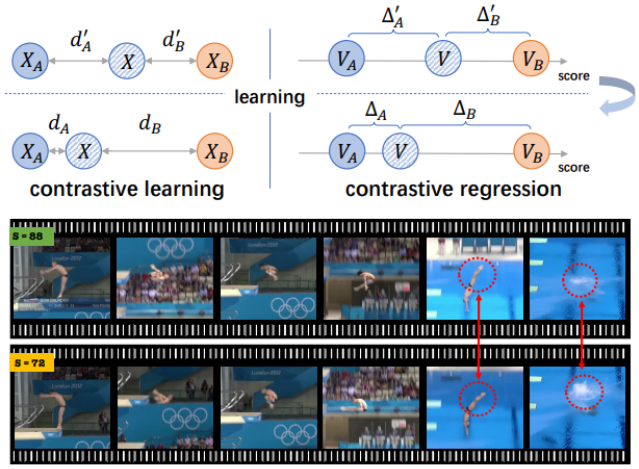


Figure 2: The Contrastive Regression (CoRe) framework for action quality assessment. Inspired by contrastive learning that learns representation by encouraging the distances of samples (e.g., d_A and d_B) to reflect their semantic relationship, the AQA model is trained to regress the relative scores (e.g., Δ_A and Δ_B) to reflect the differences of action quality among videos. By comparing two videos with different scores, CoRe encourages the model to learn from differences between videos for assessment.[15]

This paradigm marks a crucial step in the field, as it changes the nature of the supervision signal to be inherently more robust to the label noise and subjectivity found in AQA datasets.

Prerequisites: For this approach to be implemented, the model needs to be trained on relative quality differences. While this requires some form of annotation, a key advantage is that existing benchmarks with absolute scores can be readily converted into a set of pairwise comparisons for training. A critical design choice within this framework is the selection of appropriate "exemplar" videos for comparison, as this can significantly impact the model's performance.

4.4 Semi-Supervised Learning

SemSL offers a direct solution to the label scarcity problem by combining a small amount of labeled data with a large amount of unlabeled data. This is a natural fit for AQA, where unlabeled videos of actions are often abundant (e.g., on the internet), while expert-labeled examples are rare.

4.4.1 The Teacher-Reference-Student Framework. A prime example of SemSL in AQA is the work of Yun et al. [16], who proposed a Teacher-Reference-Student (TRS) architecture.

The steps are as follows:

(1) A "teacher" network is first trained on the small, expert-labeled dataset.
(2) The trained teacher then processes the large unlabeled dataset to generate "pseudo-labels" (i.e., predicted scores).

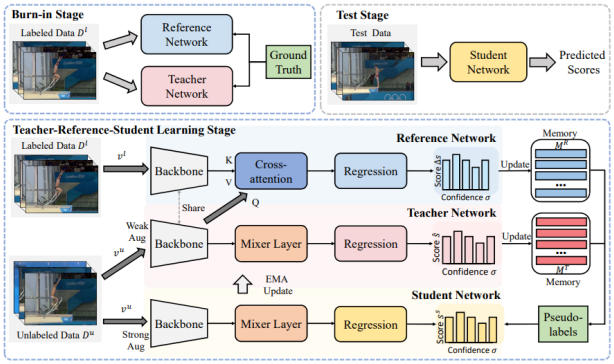


Figure 3: The overall framework of method. Method starts with a burn-in stage of training, followed by the teacher-reference-student learning stage. It mainly contains a teacher network, a student network and a reference network. The teacher network and reference network predict pseudo-labels for unlabeled data and supervise the training of the student network. The student network has the same architecture as the teacher. Moreover, confidence memory is used to ensure the reliability of pseudo-labels generated by the teacher network and reference network. [16]

(3) A "reference" network and a confidence memory module filter these pseudo-labels, retaining only the most reliable ones.
(4) A "student" network is then trained on a combined dataset of the original labeled data and the high-confidence pseudo-labeled data.

The framework operates in an iterative cycle. In each iteration, instead of the student model simply replacing the teacher, the teacher's parameters are updated via an exponential moving average (EMA) of the student's parameters. This technique allows the teacher to evolve into a more stable, time-averaged version of the student. This approach was shown to significantly improve model performance by effectively leveraging vast amounts of unlabeled video.

Prerequisites: The primary prerequisite for this approach is an initial set of labeled data. This small 'seed' dataset is essential for training the teacher model during what the authors refer to as the "burn-in stage

4.5 Self-Supervised and Unsupervised Learning

SelSL takes the reduction of supervision a step further by learning representations directly from unlabeled data, without any manual annotations.

4.5.1 The Self-Supervised Workflow in AQA. In the context of regression-based AQA, self-supervised learning typically follows a two-stage workflow:

(1) **Pre-training (Label-Free):** A model (e.g., a 3D CNN) is trained on a large, unlabeled video dataset using a pretext task. The pretext task is designed to force the model to learn features that are semantically meaningful for the domain. For example, the model might be asked to predict the temporal order of shuffled video clips, or to determine if two clips show the same person from different viewpoints. The supervision for this task is generated automatically from the data itself, requiring no human labels.
(2) **Fine-tuning (Supervised):** After pre-training, the learned feature extractor (the body of the CNN) is transferred to the downstream AQA task. A new regression head is added, and the entire model (or just the new head) is fine-tuned on a small, labeled AQA dataset.

The key idea is that the representation learned during the label-free pre-training stage provides a much better initialization for the final supervised task, allowing the model to achieve high performance with very few labeled examples.

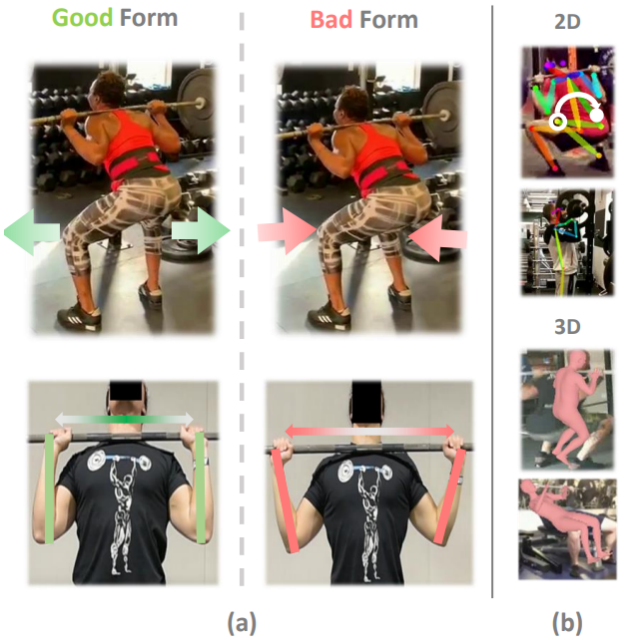


Figure 4: Concept illustration from [5]. (a) Errors of small magnitude in workout form: The 'Good' column shows correct posture, while the 'Bad' column shows erroneous form. (b) Failures of off-the-shelf 2D and 3D pose estimators in gym scenarios. The referenced work tackles the problem of detecting workout errors by replacing these error-prone estimators with more robust, fitness-oriented representations learned via self-supervision.

4.5.2 Pretext Tasks from Domain Knowledge. One powerful approach to SelSL is to leverage inherent domain knowledge to design the pretext task. Parmar et al. [5] provide a clear example of this in the context of workout form assessment. They exploit the fact that many exercises involve harmonic, cyclical motions. Their model is pre-trained on unlabeled gym videos with a pretext task of motion disentangling and pose contrastive learning, which capitalizes on the harmonic nature of the movements. This domain-informed

pre-training allows the model to learn robust pose-sensitive representations that are highly effective for the downstream task of detecting subtle errors in workout form.

4.5.3 Self-Training with Large-Scale Models. Another powerful paradigm is self-training, where a model iteratively improves by training on its own generated data. Zohar et al. [22] introduced Video-STaR, a self-training framework for adapting large vision-language models (LVLMs) to new tasks. This approach uses a pre-trained LVLM, such as Video-LLaVA, as its foundation, and does not train any model from scratch. The core of Video-STaR is an iterative cycle of generation and fine-tuning.

The filtering module in Video-STaR is a crucial component, termed the "Parser-Verifier." Its function is to ensure that only high-quality, self-generated data is used for re-training. The process is as follows:

(1) **Generation:** The LVLM is prompted to generate a textual answer for a video, which should contain the ground-truth label (e.g., an action class or a numerical score).
(2) **Parsing:** A parser, using a combination of regular expressions (for structured data like bounding boxes) and named entity recognition (for more nuanced concepts), extracts the predicted labels from the generated free-form text.
(3) **Verification:** A verifier module compares the extracted labels to the original ground-truth labels using a task-appropriate metric. For numerical scores, this could be a simple tolerance threshold (e.g., within 5% of the true score). For temporal localization, it would be Intersection over Union (IoU).
(4) **Filtering:** Only the generated question-answer pairs where the verifier confirms a match are retained for the next stage of instruction tuning.

This cycle allows the model to adapt to new domains and tasks, such as AQA, using existing supervision signals that are not in the traditional instruction-tuning format.

4.6 Leveraging External Knowledge: Vision-Language Models

The most recent and perhaps most transformative trend in label-efficient AQA involves leveraging the vast external knowledge embedded in large-scale, pre-trained foundation models.

4.6.1 The Role of CLIP in AQA. A key enabler of this trend is CLIP (Contrastive Language-Image Pre-training) [10]. CLIP is a foundation model trained on hundreds of millions of image-text pairs from the internet. Its architecture consists of two encoders: one for images and one for text. It is trained with a simple yet powerful contrastive objective: given a batch of image-text pairs, the model learns to predict which text caption corresponds to which image. This forces the model to learn a shared, multi-modal embedding space where visual and textual concepts are aligned. For example, the image of a dog and the text "a photo of a dog" will have very similar vector representations in CLIP's embedding space.

This rich, semantic alignment is highly relevant to AQA. It allows researchers to use natural language as a form of supervision. The work of Xu et al. [13] pioneers this direction by proposing a framework for Vision-Language Action Knowledge Learning. Instead of relying solely on numerical scores, their approach uses textual descriptions of actions (e.g., "a forward dive with 2.5 somersaults in pike position") as a form of supervision. By aligning video features with the rich semantic embeddings of these textual descriptions from CLIP, the model learns to understand the nuances of an action. This approach effectively replaces expensive numerical annotation with cheaper, more descriptive textual annotation.

Similarly, Zhou et al. [20] propose a framework to align the AQA task with the broader capabilities of pre-trained models. They reformulate AQA as a coarse-to-fine classification problem (e.g., classifying a performance as "fair," "good," or "excellent"), which is more aligned with the classification tasks on which foundation models are typically pre-trained. This reduces the domain shift between pre-training and fine-tuning, leading to better performance and generalization. These VLM-based approaches represent a paradigm shift, moving from learning from scratch or from in-domain data to effectively transferring and adapting vast, pre-existing knowledge for the specific task of AQA.

5 DISCUSSION AND ANALYSIS

The survey of learning paradigms reveals a clear trajectory in AQA research: a progressive movement away from dependence on dense, expert-annotated scores towards more flexible, scalable, and data-efficient methods. This section synthesizes these findings, provides a quantitative comparison, highlights persistent challenges, and outlines promising directions for future work.

5.1 Quantitative Comparative Analysis

While conceptual shifts are important, their value is ultimately measured by empirical performance. To provide a concrete comparison, we synthesized performance data from the source papers on two major AQA benchmarks: AQA-7 and MTL-AQA. We focus on the Spearman's Rank Correlation (ρ), as it is the most widely reported metric.

Table 3: Performance comparison (Spearman's ρ) on the AQA-7 dataset. Results show a general trend of improvement with more advanced, label-efficient methods. * indicates our synthesis of reported results.

Method	Paradigm	Avg. Corr. (ρ)
C3D-SVR [6]	Supervised	0.694
JRG [4]	Supervised	0.785
USDL [12]	Supervised (Uncertainty)	0.810
CoRe+GART* [15]	Contrastive	0.840
DAE-CoRe* [18]	Contrastive (Uncertainty)	**0.852**

Table 3 shows the performance progression on the diverse AQA-7 dataset. Early supervised methods like C3D-SVR established a baseline. More structured approaches like JRG (Joint Relation Graphs) and methods accounting for label uncertainty like USDL brought significant gains. However, the introduction of contrastive regression with CoRe marked a substantial leap in performance, which was further improved by incorporating uncertainty modeling in DAE-CoRe. This suggests that learning relative differences is a more effective strategy than direct regression for this diverse, multi-sport

dataset. Although some works report R-L2, we restricted our quantitative synthesis to Spearman's ρ as it is by far the most consistently reported metric across datasets, allowing for fairer comparison.

Table 4: Performance comparison (Spearman's ρ) on the MTL-AQA dataset (diving). The table shows results with and without using the Degree of Difficulty (DD) as an input feature.

Method	Paradigm	Avg. Corr. (ρ)	
		w/o DD	w/ DD
C3D-AVG-MTL [7]	Supervised (MTL)	0.904	-
USDL [12]	Supervised (Uncertainty)	0.907	0.923
MUSDL [12]	Supervised (Uncertainty)	0.916	0.927
CoRe+GART* [15]	Contrastive	0.934	0.951
DAE-CoRe* [18]	Contrastive (Uncertainty)	-	0.959

Table 4 shows a similar trend on the more homogeneous MTL-AQA diving dataset. Again, contrastive methods (CoRe, DAE-CoRe) outperform direct regression approaches (USDL, MUSDL). The results also show that leveraging additional metadata, such as the Degree of Difficulty (DD), consistently improves performance across all methods. This quantitative analysis confirms that the conceptual shift towards contrastive learning and uncertainty modeling has yielded tangible improvements in state-of-the-art performance.

Table 5: Performance comparison (Spearman's ρ) on the JIGSAWS dataset. The results demonstrate a clear trend of improvement, validating the effectiveness of contrastive and uncertainty-aware paradigms in the domain of surgical skill assessment.

Method	Paradigm	Avg. Corr. (ρ)
JRG [4]	Supervised (Structured)	0.57
MUSDL [12]	Supervised (Uncertainty)	0.70
CoRe+GART* [15]	Contrastive	0.85
DAE-CoRe* [18]	Contrastive (Uncertainty)	**0.86**

To test the generalizability of these paradigms beyond sports, we also analyzed their performance on the JIGSAWS dataset, which involves fine-grained surgical skill assessment. Table 5 illustrates a performance trajectory consistent with that observed on sports datasets. Structured supervised methods like JRG provide a strong baseline, which is significantly improved upon by methods that model label uncertainty, such as MUSDL. The transition to contrastive learning with CoRe again marks a substantial leap in performance, ultimately topped by DAE-CoRe, which combines both contrastive learning and uncertainty modeling. This trend strongly suggests that the advantages of learning relative differences are domain-agnostic and particularly effective for tasks requiring the evaluation of precise, fine-grained motor skills.

5.2 Open Challenges and Research Gaps

Despite significant progress, several key challenges remain at the forefront of AQA research.

5.2.1 Generalization and Long-Form Video. Most AQA research has focused on short, isolated actions. As highlighted by the LOGO dataset [19], assessing quality in long-form videos (several minutes) that contain multiple actors and complex temporal dependencies remains a major hurdle. Methods must not only evaluate individual actions but also understand the long-term narrative and group dynamics. Generalizing across different domains (e.g., from sports to surgery) is another significant, largely unsolved challenge.

5.2.2 Explainability, Trust, and Bias. As AQA systems are deployed in high-stakes applications like medical training or official sports judging, their "black box" nature becomes a critical liability. Most deep learning models provide a score with no explanation, making it difficult to trust their outputs or diagnose failures. The neuro-symbolic approach proposed by Okamoto and Parmar [3] offers a compelling alternative. By using neural networks to extract interpretable symbols (like pose and splash size) and then applying a transparent, rule-based system to derive a score, their method provides a fully detailed and verifiable report. This work points towards a crucial research direction: building AQA systems that are not only accurate but also transparent and trustworthy. This is essential for ensuring fairness and accountability.

5.2.3 Uncertainty Quantification. The scores used to train AQA models are often subjective averages from multiple human judges. This introduces aleatoric uncertainty—inherent noise or ambiguity in the labels themselves. Zhang et al. [18] address this directly by proposing a model that predicts a score *distribution* rather than a single point value. By explicitly modeling the uncertainty, their Distribution Auto-Encoder (DAE) can better represent the inherent ambiguity in the data and achieve a more robust fit. Further research into quantifying both aleatoric (data) and epistemic (model) uncertainty is needed to build more reliable AQA systems.

5.3 The Human in the Loop: User-Centric AQA

A critical dimension often overlooked in purely technical surveys is the user perspective. The ultimate goal of an AQA system is not just to produce a number, but to provide a useful tool for a human user. This user could be an expert (coach, surgeon, referee) or a layperson (athlete, patient, student).

For expert users, AQA systems can serve as powerful assistive tools. The work on visual analytics for movement data [1, 11] shows how interactive dashboards can help analysts explore large datasets, identify behavioral patterns, and validate hypotheses about group movements. In AQA, a similar interactive system could allow a coach to explore a diver's performance, scrubbing through the video while seeing synchronized data streams of joint angles or distances from the board. The explainable reports generated by neuro-symbolic systems [3] are a prime example of user-centric design; they provide not just a score, but visual evidence and a breakdown of errors that a judge or coach can use to inform their own assessment or provide targeted feedback to an athlete.

For lay users, the requirements are different. A patient undergoing physical therapy at home does not need a detailed biomechanical report but rather simple, real-time, and actionable feedback. The work on workout form assessment [5] is a step in this direction. Such a system could be simplified to provide binary feedback

("Good form" vs. "Knees too far forward") or simple visual cues overlaid on the video feed. The challenge here is to distill the complex output of an AQA model into an intuitive and effective user interface.

5.4 Future Directions

The confluence of label-efficient learning and the rise of foundation models points to several exciting future directions for AQA.

5.4.1 Hybrid Neuro-Symbolic Architectures. The trade-off between the performance of end-to-end deep models and the explainability of symbolic systems suggests that hybrid approaches are a promising path forward. Future systems could leverage the power of large foundation models for robust feature extraction and semantic understanding, while using a symbolic reasoning layer to ensure transparency and explainability. For example, a VLM could provide a high-level semantic description of a surgical procedure, which is then fed into a rule-based system [3] that checks for adherence to a predefined protocol, providing a fully auditable assessment. This approach could combine the semantic richness of models like CLIP with the logical rigor of symbolic AI, leading to AQA systems that are both highly accurate and fully transparent.

5.4.2 Advanced Label-Efficient Learning. There is significant room to combine the strengths of different label-efficient paradigms. One promising direction is **Uncertainty-Aware Semi-Supervision**. In a teacher-student framework [16], the student model is often trained on pseudo-labels from the teacher, but not all pseudo-labels are equally reliable. A model that can estimate its own uncertainty, such as the DAE from Zhang et al. [18], could be used as the teacher. The uncertainty estimate for each pseudo-label could then be used to weight its contribution to the loss function, effectively telling the student to pay less attention to less reliable predictions. This could lead to more stable and effective semi-supervised training, especially in noisy, real-world settings.

5.4.3 Multi-Modal AQA. Current AQA is overwhelmingly vision-based. However, many real-world scenarios involve multiple modalities. In sports, the audio from a coach's instructions or a commentator's analysis [7] provides a rich, natural source of weak supervision. In healthcare, data from wearable inertial measurement units (IMUs) can provide precise kinematic data that complements video-based pose estimation. Future AQA systems should aim to fuse these diverse data streams to build a more holistic and robust understanding of action quality. This could involve developing new fusion architectures or using one modality (e.g., text) to supervise the learning process of another (e.g., video).

6 CONCLUSION

This paper has presented a comprehensive survey of the evolving landscape of Action Quality Assessment, with a particular focus on the critical shift from fully supervised learning to a diverse array of non-supervised and label-efficient paradigms. We began by establishing the importance of AQA and outlining the fundamental limitations of the traditional supervised approach, namely its prohibitive reliance on costly, subjective, and labor-intensive data annotation.

Our survey charted a clear evolutionary path, structured by a taxonomy based on the degree of supervision. We reviewed foundational supervised methods and then detailed the emergence of more robust and scalable paradigms, including contrastive learning, semi-supervised learning, self-supervised learning, and vision-language models. Our quantitative analysis confirmed that these conceptual shifts have yielded significant performance gains on standard benchmarks.

However, significant challenges remain. The pursuit of AQA systems that are not only accurate but also generalizable, explainable, and trustworthy is an ongoing endeavor. We have highlighted the importance of tackling long-form, multi-person scenarios, the need for transparent and interpretable models as exemplified by neuro-symbolic approaches, and the necessity of designing systems with the human user in mind. The future of AQA is bright and will likely be shaped by the continued integration of large-scale foundation models, the development of hybrid reasoning systems, and a steadfast focus on creating technologies that are not only powerful but also accessible, fair, and useful in real-world applications. We hope this survey provides a valuable resource and a clear roadmap for researchers and practitioners working to advance this important and impactful field.

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